

Introduction

If there's one thing we can say with absolutely certainty the human race has been striving towards since the time we started thinking is this – finding an easier way to do something. Anything. Be it getting from your home to school, managing your shopping, trying to crack the code on 100% earthquake resistant building material, or making adorable cat videos. To do all of that (and more) better than we've ever done before, utilizing the Internet is key.

The Internet has not just democratized knowledge in the 21st century, it has also become an important pivot on which our species' progress depends heavily, playing an important catalytic role in anything we do or achieve from this moment on.

We all know how the Internet literally spreads across the globe much like a spider's web, with every node in its interwoven, multilayered structure representing a computing device with a unique IP address. We have a few billion people connected to the web right now – think about the computing devices they own. There's more. Add everything from your watch, car, mixer, oven, washing machine, fridge, thermostat, light bulbs, flower pot, clothes, sunglasses, door lock, and fifty different household items to that list. That's just the number of devices which are going to come alive, thanks to the Internet. But what about the devices that power the Internet itself? If you put all of this together, there's only one number that anyone can come up with – mindboggling. And that's just the start of the Internet of Things.

With a myriad sensors interconnected on a vast global network, we can learn and track processes (internal or external) to make better, informed decisions. There's no hit and miss anymore, only tried and tested behaviour models to learn from with the help of machines, putting us on an exponential path of progress like never before. If the Industrial Revolution led to the events that put a man on the moon, then just try and imagine